

responsiveness and high frequency bi-directional connectivity, etc.

Commercial IoT platforms are classified into five sections.

(i) **End-to-end IoT platforms:**

These are full-stack platforms performing all services, including connectivity, protocol interoperability, data-structure interoperability, device management, device firmware/software upgrade management, rules engine, actions engine, ticketing management, data visualisation, analytics, multiple applications and accounts, applications templates, applications reuse and duplication, user management, dashboard building and customisation, reports, external interfaces, security, and data storage.

(ii) **Connectivity management platforms:**

These are elementary IoT platforms that offer solutions for connectivity orchestration, connection management and provisioning of communication services for connected IoT devices.

(iii) **Cloud storage platforms:** Cloud based IoT platforms are the measurable enterprise grade back-end for data management of IoT applications.

(iv) **Device management platforms:**

These offer device provisioning, remote administration, software updation, firmware upgrades, security patching, remote control of devices, monitoring and reporting about specific metrics associated with IoT assets.

(v) **Analytics platforms:** Analytics infrastructure runs analytical applications, Big Data processing, contextual mining and machine learning algorithms. Analytics platforms provide statistical and optimisation tools to extract valuable information from structured and unstructured data.

Here is a list of a few popular IoT platforms:

- Amazon AWS IoT
- Google Cloud IoT
- Microsoft Azure IoT
- Oracle IoT Cloud
- IBM Watson IoT

Open standards and a horizontal architecture

Developing open guidelines enables a horizontal architecture for IoT arrangement suppliers and gadget providers to expand on. A standard, IoT industrywide design aids in similarity and start-to-finish coordination since arrangement suppliers, IT associations and gadget makers can anticipate regular prerequisites when creating and actualising arrangements.

Horizontal architecture provides open connectivity, improved security, scalability and flexibility. It provides end-to-end integration, more solutions (i.e., normalisation, analytics and streaming), better privacy and security, higher quality and lower cost. IT organisations, in particular, are at the heart of enterprise IoT implementations and can help lead the transition to standards-based horizontal architecture.

Different horizontal platforms

Canopy

Canopy simplifies the IoT cloud by acting as a cloud relay between IoT devices and applications. It enables rapid development of IoT solutions for consumer, commercial and industrial markets. Canopy is a collection of software tools and libraries that simplify the development of cloud-enabled products. By 'cloud-enabled product', we mean any physical device with a connection to the Internet that can be monitored and/or controlled over the Web. Such products may include consumer appliances, industrial equipment and other IoT devices. A cloud-enabled product needs software

to run in several places -- on the device, in the cloud and within client-side applications. Canopy contains sub-projects for each of these:

- Canopy Embedded Development Kit (EDK) software runs on the device.
- Canopy Cloud Service (CCS) software runs on the server-side.
- Canopy Application Development Kit (ADK) software runs within an application — for example, on a smartphone or in a Web browser.

Chimera IoT

Chimera IoT provides real value to the world through IoT using anonymised data, crowd sourced remediation, and sophisticated rules based analytics integrated with AI and validated through a human feedback loop. Chimera IoT uses six technology products and services including HTML5, Google Analytics and Google Fonts. The six technologies it uses for its website include Amazon, Google and Sectigo SSL.

DeviceHive

DeviceHive is free open source software that helps in smart gadgets communication and its management. It comprises a communications layer and control software with multi-platform libraries, which allow users to improve smart energy, robotisation, remote sensing, remote control, telemetry, and more. DeviceHive also provides REST, Web socket APIs by default plus MQTT API as an extra plugin. All communication is performed by means of JSON messages.

Nimbits

Nimbits is a hybrid cloud server. This horizontal platform is built on constrained embedded systems by suggesting IoT related services. It can run on Amazon EC2, Google App Engine, Ubuntu Linux KVM based virtual machine and also